

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	ROW Geomatics Survey Report Review
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ROW Geomatics Survey Report Review

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Revision History

Rev No.	Date	Description of Revision
R00	05 Aug 2022	Issued for Interdisciplinary Checks
R01	08 Aug 2022	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved “As-Built” documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the “As-Built” document to Z02).

Restricted

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-001

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

Abbreviations

Abbreviation	Meaning
BFPCL	Brass Fertilizer Petrochemical Company Limited
TCE	Tata Consulting Engineers
GPMC	Gass Processing & Methanol Complex
GGPL	Gass Gathering Pipelines
FEED	Front End Engineering Design
ROW	Right of Way
MTPD	Metric Tonnes Per Day
TQ	Technical Query
BBOU	Bubouwe Bou
ELEP	Elepa
NUNR	Nun River
SEIB	Seibou
OPUG	Opugbene
LGA	Local Government Area
DEP	Design and Engineering Practices

Table of Contents

Abbreviations	4
1. INTRODUCTION	6
1.1. Background Information	6
1.2. Purpose	7
2. REGULATIONS, CODES AND STANDARDS	7
3. PIPELINE ROUTE RIGHT OF WAY	8
4. PARCELLATION	10
5. DESCRIPTION OF REVIEW	11

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

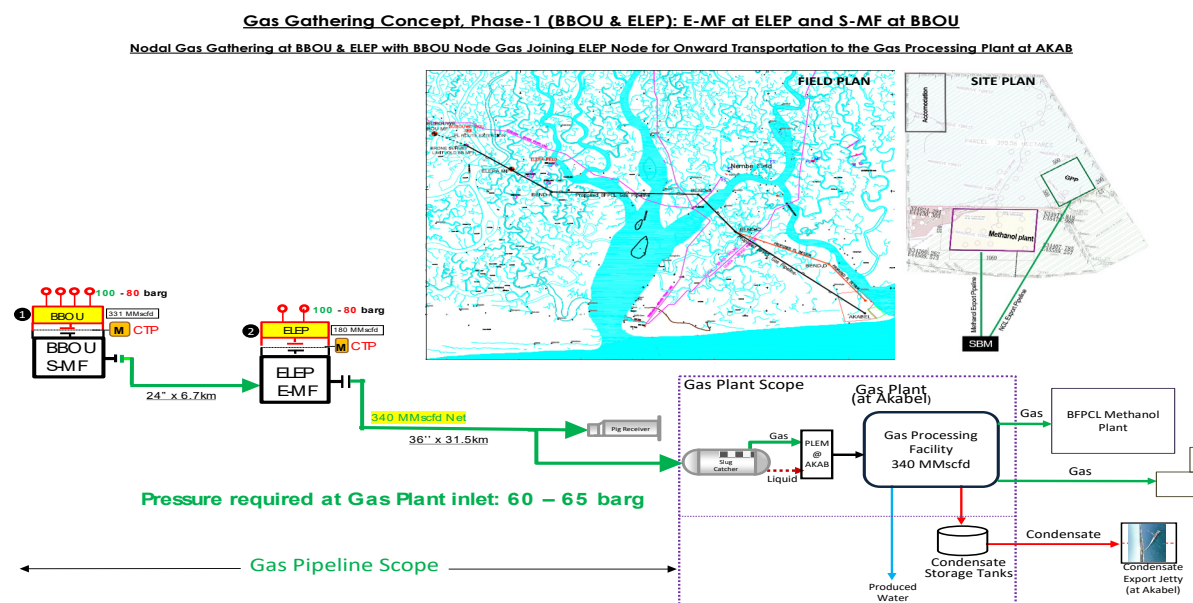


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of this document is to review the adequacy or otherwise of the Geomatics survey as carried out by M/s BRONE

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Pipeline Route Right of Way

The observations include the following as a minimum;

- The survey as carried out traversed from the methanol plant location at Akabel to the 'old' BB manifold
- SPDC indicated a new location with coordinates for the BBOU manifold and the survey right of way has not captured the changes to the new manifold location
- The political sensitivity around the Otokolopiri area makes it imperative to re-route pipeline from Fantuo community (Bend C) to land at GPP area (See Figure 2 below).

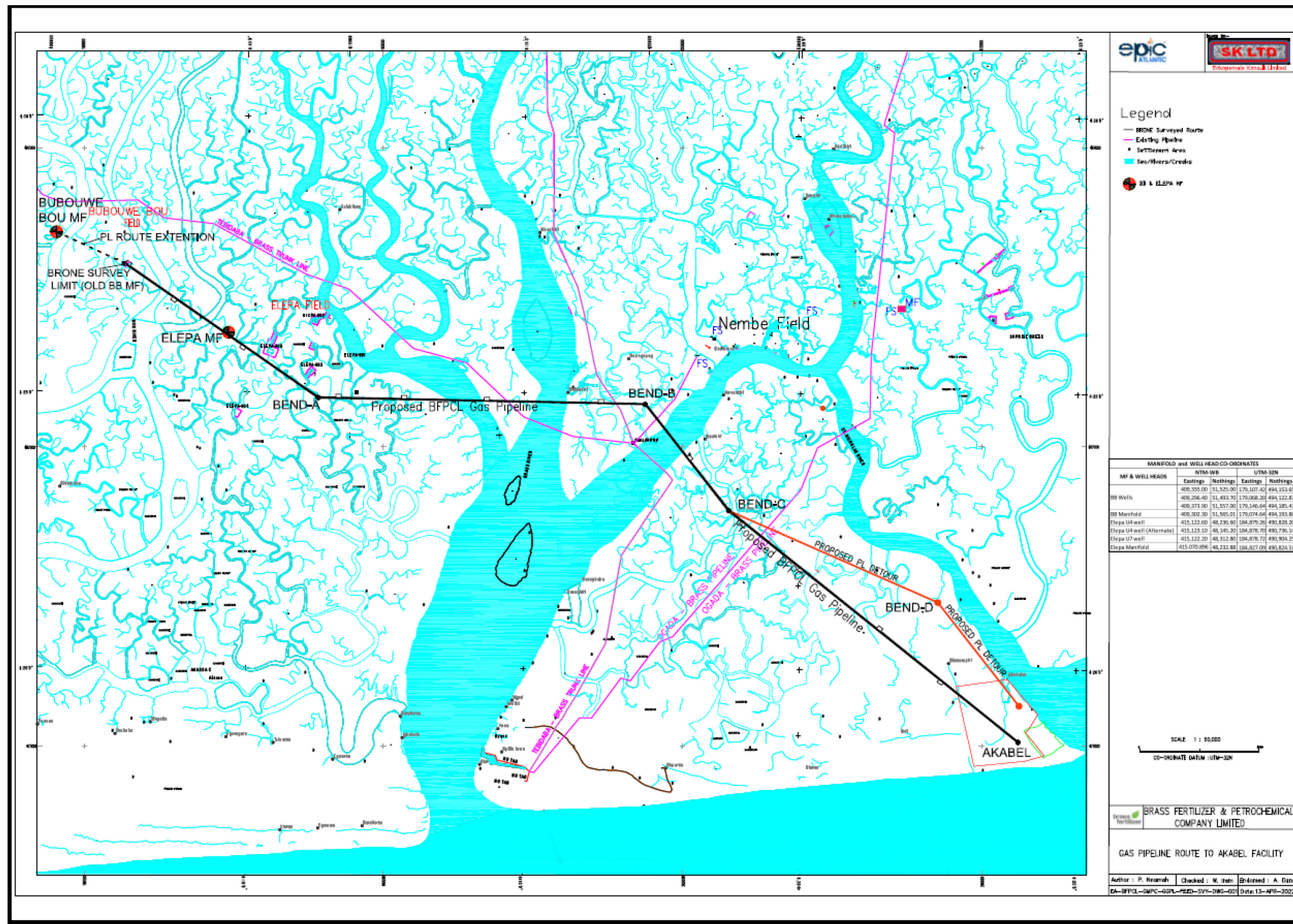


Figure 2: Pipeline Route

4. Parcellation

The parcellation survey did not precisely structure ownership communities on the Right of Way more so as the new 2.5 km extension of the route is yet to be surveyed, and re-routing at the Akabel end is being considered, update to ownership of land has to be made.

5. Description of Review

The following were observed.

- Pipeline RoW description conventionally is from source of flow, that is from BBOU to Akabel
- The bathymetric data of some creeks and creek-lets were not indicated on the drawing.
- All rivers, creeks and creek-lets should be labelled serially end-to-end with appropriate IDs for easy referencing.
- Bathymetric chart datum should be tied to project elevation datum and not on predicted tide.
- It's reported that topographical spot-heightening was conducted on Right of Way centreline at 25m interval, but the centreline was never traversed.
- There is no topographical elevation data on the entire Right of Way.

RECOMMENDATION

1. A pre-Engineering route survey will have to be carried out for the RoW to capture all identified missing information, based on which the pipe flow hydraulics analysis will be repeated.
2. The FEED will therefore have to be updated based on the results of the pre-engineering surveys.